AI CAREERS FOR WOMEN: LINKEDIN FELLOWSHIP

Project Title: FeedbackForward PG — Postgraduate-Specific Teaching Intelligence System

Project Domain: Educational Analytics & NLP — Excel + Power BI + Copilot/Chatbot

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3. Problem Statement

Organizations across higher education collect massive volumes of data daily, but this raw information is frequently inconsistent or poorly structured, making it exceptionally difficult to derive actionable insights. At National Post Graduate College, student feedback often remains underutilized because decision-makers lack the specialized tools and technical skills required to process qualitative data effectively.

Furthermore, several critical innovation gaps remain unresolved in the current project:

- **Postgraduate-Specific Dynamics:** Treating PG and UG feedback with the same analysis framework produces misleading conclusions, as PG students have deeper subject knowledge and different faculty expectations regarding research expertise.
- **Response Rate Distortion:** Typical response rates of 30–40% significantly bias all other KPIs, yet these analytical distortions are currently unhandled.
- **Skills-Research Mismatch:** There is no existing analysis that cross-references student feedback on "research exposure" or "subject currency" with actual faculty research profiles.
- **Departmental Culture Variation:** A single "Common Themes" KPI across all departments may mask important departmental-level patterns in feedback behavior and tone.
- **AI Quality Assurance:** The current AI outcomes lack a framework for evaluating whether generated summaries are accurate, free of hallucinations, or actually useful for faculty decisions.

4. Objectives

The specific, measurable goals for this project include:

- **Data Exploration & Cleaning:** To systematically collect raw student feedback and conduct an audit to handle missing entries and duplicates using advanced Excel and Power Query.

- **PG-Specific Analysis Module:** To develop separate benchmarks and KPI thresholds that account for the higher critical expectations and research orientation of postgraduate students.
- **Response Rate Bias Correction:** To implement statistical sampling weights that adjust KPI scores to reflect the likely views of non-respondents, improving analytical validity.
- **Faculty Research-Feedback Alignment Index:** To cross-reference student feedback on 'subject currency' with actual faculty research output (publications, projects) to identify genuine development needs.
- **Departmental Subgroup Analysis:** To conduct department-wise subgroup analysis with separate feedback culture calibration for humanities, sciences, and commerce streams.
- **AI Prompt QA Protocol:** To develop a protocol for the AI prompt library that includes accuracy checking, hallucination detection, and usefulness rating steps.
- **Visual Insight Delivery:** To build a high-fidelity interactive dashboard featuring KPI cards and slicers that allow stakeholders to filter data by department or semester.

Core Analytical KPIs to be Monitored

KPI Name	Description	Tools Used
Sentiment Score	Measures the tone of student text (Positive/Neutral/Negative).	Microsoft Copilot
Average Rating	Calculates the mean numerical score across various teaching parameters.	Power BI DAX
Bias-Corrected Response Rate	Tracks participation levels and applies inverse probability weighting to adjust scores based on non-respondent likelihood.	Excel Formulas / Power BI
Common Themes	Identifies recurring phrases like "Course Material" or "Class Clarity".	AI NLP
Research-Feedback Alignment Index	Cross-references student "subject currency" needs with actual faculty publication data.	Power BI / Excel
Departmental Normalized Score	Calibrates scores to account for varying feedback cultures across different academic streams.	Power BI DAX

5. Methodology/Implementation Plan

This project follows an enhanced five-phase workflow:

- **Phase 1 — PG Benchmark Calibration:** Sourcing raw datasets and researching PG feedback norms from NAAC/IQAC literature to define appropriate KPI thresholds distinct from UG norms.
- **Phase 2 — Preprocessing & Bias Correction:** Using Excel's "Text-to-Columns" and "IFERROR" for hygiene, while applying inverse probability weighting to re-weight respondent scores.

- **Phase 3 — Research-Feedback Alignment:** Applying Power BI measures and collecting faculty publication data to compute a simple alignment score visualized as a research-feedback gap matrix.
- **Phase 4 — Visualization & Normalization:** Constructing interactive dashboards that segment all KPIs by department and apply statistical normalization within each department before comparison.
- **Phase 5 — Documentation & QA:** Finalizing the project report and designing a 5-question evaluation rubric for each AI output to ensure accuracy, actionability, and completeness.

6. Resources and Timeline

- **Hardware:** Laptop with Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended for Power BI).

- **Software:** Microsoft Excel (365/2019+), Power BI Desktop, and Microsoft Copilot or web-based AI Chatbots.

- **Dataset:** Raw CSV/Excel data obtained from Kaggle, UCI Repository, or Edunet-provided sources.

Project Timeline:

- **Week 1:** Dataset collection, exploration, and PG norm research.
- **Week 2:** Data cleaning, bias correction, and prompt design.
- **Week 3:** Analysis, AI tool integration, and dashboard/output build.

- **Week 4:** QA testing, report writing, GitHub upload, and video submission.

7. Expected Outcomes

- **India's First PG-Specific Teaching Feedback Framework:** Featuring calibrated norms for higher education assessment in Indian PG colleges.
- **Statistically Bias-Corrected Dataset:** Demonstrating advanced analytical thinking that improves the validity of all downstream insights.
- **Research-Feedback Alignment Report:** Identifying where faculty development investment is most needed, directly actionable by the Principal.
- **Validated AI Prompt Library with Quality Scores:** A reusable, reliable resource that other NPGC faculty can immediately use, extending the project's impact.

8. References

- **Edunet Foundation:** AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org.

- **Edunet Foundation (YouTube):** SkillsBuild and Next Gen Employability Tutorials.

- **LinkedIn Learning:** AI and Data Analytics Courses. 2026.
- **Microsoft Corporation:** Microsoft Power BI and Excel Documentation.
- **Microsoft Corporation:** Microsoft Copilot.
- **Kaggle Inc:** Open Datasets.
- **UNESCO:** AI in Higher Education: Emerging Trends and Policy Frameworks.
- **Harvard Business Review:** Using AI to Improve Organizational Feedback Systems.
- **GitHub Guides:** Best Practices for Documenting Data Science Projects.
- **NAAC/IQAC Literature:** Guidelines for Postgraduate institutional benchmarking.